***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 3 Microbial Metabolism

3.1 Multiple Choice Questions

1) The prokaryotic transport system that involves a substrate-binding protein, a membrane-integrated transporter, and an ATP-hydrolyzing protein is
A) the ABC transport system.
B) group translocation.
C) symport.
D) simple transport.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.2

2) The sum of all biosynthetic reactions in a cell is known as
A) metabolism.
B) anabolism.
C) catabolism.
D) synthatabolism.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.1

3) Based on the functional roles of phosphate in various microbial metabolisms, which of the following compounds most likely contain phosphate?
A) organic compounds
B) inorganic compounds
C) both organic and inorganic compounds
D) neither organic nor inorganic compounds
Answer: C
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.1

4) Which of the following would be used by a chemoorganotroph for energy?
A) $C_2H_3O_2^-$
B) H_2
C) CO_2
D) H^+
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.3

5) Which of the following statements is FALSE?
A) Most bacteria are capable of using ammonia as their sole nitrogen source.
B) Some bacteria are able to use nitrates or nitrogen gas as their nitrogen source.
C) Most available nitrogen is in organic forms.
D) Nitrogen is a major component of proteins and nucleic acids.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.6

6) All microorganisms require
A) carbon, iron, and sodium.
B) phosphorus, aluminum, and sodium.
C) calcium, potassium, and magnesium.
D) phosphorus, selenium, and sulfur.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.1

7) Which element functions BOTH as an enzyme cofactor and as a stabilizer of ribosomes and nucleic acids?
A) iron
B) hydrogen
C) zinc
D) magnesium
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.1

8) Based on your understanding of metabolism, generalize when an enzyme′s rate of activity can be changed.
A) before enzyme production
B) during enzyme production
C) after enzyme production
D) at any point—before, during, or after enzyme production
Answer: C
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.5

9) The change in Gibbs free energy for a particular reaction is most useful in determining
A) the amount of energy catalysts required for biosynthesis or catabolism.
B) the potential metabolic reaction rate.
C) whether there will be a requirement or production of energy.
D) energy stored in each compound.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.4

10) Which is an example of a micronutrient?
A) arginine
B) inorganic phosphorous
C) iron
D) vitamin B12
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.1

11) Aseptic technique refers to
A) the microbial inoculum placed into a test tube or onto a Petri plate.
B) a series of practices to avoid contamination.
C) the autoclave and other sterilizing procedures.
D) cleanliness in the laboratory.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 1.9

12) To ensure growth of a newly discovered bacterium with unknown nutritional requirements, it would be best to begin with a ________ medium rather than a ________ medium.
A) complex / minimal
B) minimal / complex
C) selective / complex
D) selective / differential
Answer: A
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.2

13) If $\Delta G^{0\prime}$ is negative, the reaction is
A) exergonic and requires the input of energy.
B) endergonic and requires the input of energy.
C) exergonic and energy will be released.
D) endergonic and energy will be released.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.4

14) Activation energy is the energy
A) required for a chemical reaction to begin.
B) given off as the products in a chemical reaction are formed.
C) absorbed as $\Delta G^{0\prime}$ moves from negative to positive.
D) needed by an enzyme to catalyze a reaction without coenzymes.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.5

15) A catalyst
A) requires more reactants but makes the reaction rate faster.
B) increases the amount of reactants produced but does not change the rate.
C) changes the rate of the reaction but does not change the end amount of products.
D) changes both the rate of a reaction and the amount of the product that will be obtained as the reaction is completed.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.5

16) The portion of an enzyme to which substrates bind is referred to as the
A) substrate complex.
B) active site.
C) catalytic site.
D) junction of van der Waals forces.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.5

17) What is the difference between a coenzyme and a prosthetic group?
A) Coenzymes are essential for an enzyme′s function and prosthetic groups only enhance its reaction rate.
B) Coenzymes are weakly bound whereas prosthetic groups are strongly bound to their respective enzymes.
C) Coenzymes are organic cofactors and prosthetic groups are inorganic cofactors.
D) Coenzymes require additional ions to bind to enzymes but prosthetic groups are able to directly interact with enzymes.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.5

18) If an oxidation reaction occurs
A) simultaneous reduction of a different compound will also occur, because electrons do not generally exist alone in solution.
B) another oxidation reaction will occur for a complete reaction, because one oxidation event is considered a half reaction.
C) a cell is undergoing aerobic respiration, because oxygen is being used.
D) a reduction reaction would not occur, because they are opposite reaction mechanisms.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.6

19) The class of macromolecules in microorganisms that contributes most to biomass is
A) carbohydrates.
B) DNA.
C) lipids.
D) proteins.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.1

20) A chemoorganotroph and a chemolithotroph in the same environment would NOT compete for
A) oxygen.
B) carbon.
C) nitrogen.
D) phosphorous.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.3

21) A chemoorganotroph and a photoautotroph in the same environment would NOT compete for
A) oxygen.
B) carbon.
C) nitrogen.
D) carbon and oxygen.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.3

22) The Embden-Meyerhof-Parnas pathway is another name for
A) the citric acid cycle.
B) glycolysis.
C) electron transport.
D) NADH production.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.8

23) The net gain of ATP per molecule of glucose fermented is
A) 1.
B) 2.
C) 4.
D) 8.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.8

24) Which of the following is a common energy storage polymer in microorganisms?
A) acetyl~S-CoA
B) glycogen
C) adenosine triphosphate
D) H_2
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.8

25) Fermentation has a relatively low ATP yield compared to aerobic respiration because
A) more reducing equivalents are used for anaerobic catabolism.
B) less ATP is consumed during the first stage of aerobic catabolism.
C) oxidative phosphorylation yields a lot of ATP.
D) substrate-level phosphorylation yields a lot of ATP.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.8

26) From the standpoint of fermentative microorganisms, the crucial product in glycolysis is
A) ATP and regenerated NAD^+; the fermentation products are waste products.
B) ethanol or lactate; ATP is a waste product.
C) CO_2; ATP is a waste product.
D) not relevant because glycolysis is not a major pathway.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.8

27) In aerobic respiration, the final electron acceptor is
A) hydrogen.
B) oxygen.
C) water.
D) ATP.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.10

28) Which of the following is NOT membrane-associated?
A) NADH dehydrogenases
B) flavoproteins
C) cytochromes
D) Cytochromes, flavoproteins, and NADH dehydrogenases all can be membrane-associated.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.10

29) During electron transport reactions
A) OH^- accumulates on the outside of the membrane while H^+ accumulates on the inside.
B) OH^- accumulates on the inside of the membrane while H^+ accumulates on the outside.
C) both OH^- and H^+ accumulate on the inside of the membrane.
D) both OH^- and H^+ accumulate on the outside of the membrane.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.11

30) The rising of bread dough is the result of
A) biotin production.
B) carbon dioxide produced by fermentation.
C) oxidative phosphorylation.
D) oxygen being released.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.8

31) Which intermediate compound(s) in the citric acid cycle is/are often used for biosynthetic pathways as well as carbon catabolism?
A) only α-ketoglutarate
B) only oxaloacetate
C) only succinyl-CoA
D) α-ketoglutarate, oxaloacetate, and succinyl-CoA
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.9

32) Microbial growth on the two-carbon acetate substrate invokes
A) the citric acid cycle for aerobic catabolism.
B) both the citric acid and glyoxylate pathways.
C) the glyoxylate pathway.
D) the glyoxylate and glycolysis pathways.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.9

33) Which is one major difference between anaerobic and aerobic respiration?
A) electron donor
B) electron acceptor
C) use of electron transport
D) use of proton motive force
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.12

34) For a carbon source, chemoorganotrophs generally use compounds such as
A) acetate, succinate, and glucose.
B) bicarbonate and carbon dioxide.
C) nitrate and nitrite.
D) acetate, bicarbonate, and nitrate.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.3

35) All of the following are non-protein electron carriers EXCEPT
A) $FADH_2$.
B) $FMNH_2$.
C) cytochromes.
D) quinones.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.10

36) Which two metabolic processes are most dissimilar?
A) citric acid cycle and glycolysis
B) glycolysis and gluconeogenesis
C) proton motive force and substrate-level phosphorylation
D) pentose phosphate pathway and glycolysis
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.13

37) How does the proton motive force lead to production of ATP?
A) ATPase requires one proton to make one ATP.
B) Protons must be pumped against a concentration gradient from outside of the cell into the cell to rotate the F_0 subunit of ATPase for the F_1 subunit to make ATP.
C) Oxidative phosphorylation of ADP by ATP synthase requires protons as cofactors in the reaction.
D) Translocation of three to four protons drives the F_0 component of ATPase which in turn phosphorylates one ADP into ATP.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.11

38) Five-carbon sugars are used in the
A) biosynthesis of DNA and RNA.
B) catabolic pentose phosphate pathway for carbon and energy.
C) biosynthesis of DNA and RNA as well as catabolic pentose phosphate pathway.
D) activation of pentoses to form glycogen for energy storage.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.13

39) Improperly functioning acyl carrier proteins (ACPs) would likely result in
A) a physiological shift to anaerobic metabolism where an energized membrane is less important for energy production.
B) enhanced growth of a bacterium due to faster growth substrate uptake by a weakened membrane.
C) no harm to bacteria, because only archaeons and eukaryotes use ACPs for fatty acid biosynthesis.
D) death for a bacterium due to poor lipid bilayer integrity.
Answer: D
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.15

40) A bacterium running low on NADPH could ________ to generate more of this coenzyme.
A) degrade an amino acid or nucleic acid
B) invoke the pentose phosphate pathway
C) degrade a fatty acid
D) use a broad specificity phosphatase with inorganic phosphatase and NADH
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.13

41) One example of an electron acceptor that can be used in anaerobic respiration is
A) NADH.
B) water.
C) nitrate.
D) FMN.
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.12

42) When culturing a chemoorganoheterophic bacterium, what outcome is LEAST likely to occur if ammonia and phosphate are provided at equal concentrations?
A) Cells require much less P to grow than N, so extra P will be used for ATP synthesis and result in a faster growth rate.
B) Cells will never consume all of the phosphate, because N is needed in higher quantities than P.
C) The final biomass of cells will be no different than if only 50% of the phosphate was provided.
D) The bacteria will import all of the ammonia to use for biosynthetic pathways.
Answer: A
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.3

43) Most of the carbon in amino acid biosynthesis comes from
A) citric acid cycle intermediates.
B) citric acid cycle intermediates and glycolysis products.
C) glycolysis products.
D) glycolysis intermediates and products.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.14

44) Which metabolic cycle or pathway is LEAST likely to be invoked during the biosynthesis of DNA?
A) citric acid cycle
B) glycolysis
C) gluconeogenesis
D) pentose phosphate pathway
Answer: C
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.14

45) Hypothetically, if free electrons existed in sufficient numbers for enzymes to use in metabolic reactions
A) a higher diversity of cytochromes would likely be observed.
B) cytochromes would be unnecessary for cells and quinones would be more important.
C) Q-cycle reactions would no longer be necessary for electron transport, but the proton motive force would otherwise be unchanged.
D) most metabolic pathways for both anabolism and catabolism would have to be rewritten.
Answer: D
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.11

46) Which metabolic strategy does NOT invoke the proton motive force for energy conservation?
A) aerobic catabolism
B) fermentation
C) chemoorganotrophy
D) photoautrophy
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.12

3.2   True/False Questions

1) ATP-binding cassette transport systems have high substrate affinity and thus help microorganisms survive in low nutrient environments.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.2

2) A bacterial isolate that grows better on a nutrient agar plate supplemented with amino acids but still grows in a nutrient agar plate lacking amino acids suggests amino acids are trace nutrients for the isolate.
Answer: FALSE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.1

3) Regeneration of oxaloacetate is essential for the citric acid cycle to be cyclical.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.9

4) Depending on the particular metabolism of a bacterium, electron transport can be used to energize and rotate ATP synthase.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.11

5) Each amino acid made during protein biosynthesis first requires a separate biosynthetic pathway to be invoked by a cell.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.14

6) The terminating step of moving electrons onto oxygen releases additional ATP during aerobic metabolism not made during anaerobic growth.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.11

7) Nitrogenases not only reduce N_2 but also can act on acetylene (C_2H_2).
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.1

8) Due to the number of phosphate groups, ATP has approximately three times more energy stored than AMP, and ADP has approximately two-thirds the energy stored of ATP.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.7

9) In a given chemical reaction, if the free energy of formation is known for all of the reactants and each of the products, the change in free energy can be calculated for the reaction.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.4

10) Free-energy calculations are dependent on the rates of the reactions.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.4

11) With respect to nitrogen utilization, relatively few bacteria can use NH_3 whereas many more can make use of N_2.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.14

12) The proton motive force is most often generated by splitting of H_2.
Answer: FALSE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.11

13) Biosynthesis of glucose can occur by compounds other than sugars via gluconeogenesis.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.13

14) If a substance is reduced, it gains electrons.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.6

15) Molybdenum is a cofactor for nitrogenase, which means every nitrogen-fixing microorganisms will not be able to fix nitrogen without Mo.
Answer: TRUE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.1

16) Magnesium is not considered a growth factor for microorganisms, because growth factors are always organic compounds.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.1

17) Cells require iron supplemented in their growth medium as a trace metal, because it is consumed by quinones during electron transport for ATP production.
Answer: FALSE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.10

18) Varied coenzyme availability increases the diversity of enzymatic reactions in both biosynthetic and catabolic pathways possible in a cell.
Answer: TRUE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.5

19) The energy released from the hydrolysis of coenzyme A is conserved in the synthesis of ATP.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.7

20) In substrate-level phosphorylation, ATP storage is depleted during the steps in catabolism of the fermentable compounds.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.8

21) Catabolic pathways are essential for microorganisms to obtain energy, because biosynthetic reactions for cellular growth generally require energy input.
Answer: TRUE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.13

22) In electron transport systems, the electron carriers are membrane associated.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.11

23) Heme prosthetic groups are involved in electron transfer with quinones.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.5

24) During the electron transport process, protons and electrons become physically separated in the cell membrane.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.10

25) Many defined growth media that support microbial growth lack malonate, which is an important precursor for biosynthesis of lipid membranes. Based on this, we can infer cells also must have a metabolic pathway to generate malonate from other compounds.
Answer: TRUE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.15

26) The net result of electron transport is the generation of a pH gradient and an electrochemical potential across the membrane.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.11

27) A bacterium that lacks an arginine biosynthetic pathway would still be able to make proteins with arginine and grow only if arginine is supplemented into the growth medium.
Answer: TRUE
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.14

3.3 Essay Questions

1) Why is energy required for nutrient transport? Give an example of a system that transports nutrients and describe what source of energy is used to move the nutrients into the cell.
Answer: Energy is required for nutrient transport because nutrient concentration outside of the cell is lower than the nutrient concentrations inside the cell, thus nutrient transport moves solutes against a concentration gradient and requires energy. There are three examples in the text. The student could describe any one of them. They are (i) Simple transporter such as lac permease. Each nutrient molecule is cotransported into the cell with a H+ ion, thus the proton motive force provides the energy to transport nutrients. (ii) Group translocation such as sugar phosphotransferases. Each nutrient molecule is modified during the transport process. The modification, in this case, phosphorylation, releases energy, thus the energy source is an energy-rich compound such as phosphoenol pyruvate or some other phosphorylated compound. (iii) ABC transporters. In this example specific binding proteins bind to nutrient molecules with high affinity. Movement of the nutrient into the cell is coupled to ATP hydrolysis, thus ATP is the source of energy for transporting nutrients.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.2

2) Explain the differences between symporters, and antiporters.
Answer: Answers should highlight differences in transport direction and energy input.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.2

3) Compare and contrast defined media and complex media. Use specific examples in your answer.
Answer: Defined media are prepared by adding individual "pure" chemicals in known quantities. In this way, the medium itself can be explicitly defined. For example, 5 mM NaCl, 3 mM KH_2PO_4, 1.5 mM NH_4Cl, 2.5% glucose, and 3% acetate is a defined medium, because each ingredient added is at a known concentration and the chemicals present are known. Complex media needs only to contain one undefined product to be considered complex or undefined. An example of an undefined medium is 5 mM NaCl, 2.5% tryptone and 2.5% yeast extract, because both tryptone and yeast extract are not individual chemical structures but instead contain an assortment of compounds at unknown (imprecise) quantities.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.1

4) Categorize the circumstances under which the same substance (molecule) can be either an electron donor or an electron acceptor.
Answer: Answers should explain that not all molecules are strictly one or the other, and each molecule must be compared to the other in a pair to determine which is the electron acceptor and which is an electron donor.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.6

5) Contrast fermentation and respiration in terms of electron donor, electron acceptor, type of ATP production, and relative number of ATP produced.
Answer: Respiration should be distinguished as using separate electron donors and acceptors (such as organic carbon as the electron donor and oxygen as the electron acceptor), while fermentation splits organic molecules in order to oxidize one part of the molecule and reduce the other part in order to regenerate NAD+. Fermentation uses substrate level phosphorylation to generate relatively few ATP, while respiration uses oxidative phosphorylation to generate more ATP.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.8

6) Summarize the roles the proton motive force has in microbial metabolism.
Answer: The proton motive force uses an energized cell membrane for ATP synthesis via ATPase, transporting some ions and molecules into and out of the cell, and flagellar rotation.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.11

7) Discuss why energy yield in an organism undergoing anaerobic respiration is less than that of an organism undergoing aerobic respiration.
Answer: One possible explanation could point to the substrate-level phosphorylation process itself as being less energy yielding than (aerobic) oxidative phosphorylation. Another reason is the fate of pyruvate itself, where fermentation is unable to take it through the higher energy yielding process, which requires O_2 as a terminal electron acceptor. Other answers could discuss the $E_0{}'$ being greatest with the O_2/H_2O redox couple in aerobic metabolism compared to anaerobic redox couples.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.12

8) Explain the biosynthetic and bioenergetic roles of the citric acid cycle.
Answer: Some of the molecules generated during the citric acid cycle, such as alpha-ketoglutarate, oxalacetate, and succinyl-CoA, can serve as precursors for the biosynthesis of critical cellular components such as amino acids, chlorophyll, and cytochromes. The bioenergetic component of the cycle should be described in the context of $FADH_2$ and NADH electron donors storing energy potential, usable in electron transport where O_2 is reduced to water.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.8

9) In an aquatic microbial community where a photoautotroph, chemoorganoheterotroph, and nitrogen fixing bacterium are present, predict an environmental perturbation that would cause only one to be outcompeted by the other two groups and explain how each group would respond.
Answer: Answers will vary but should highlight a unique feature of one of the groups such as: photoautotrophs are sensitive to photon (light) availability, chemoorganoheterotrophs require organic molecules for carbon, and nitrogen fixing bacteria use N_2 gas.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.3

10) Differentiate between exergonic and endergonic in terms of free-energy calculations.
Answer: A positive change in free energy ($\Delta G^{0\prime}$) means the reaction needs energy input to occur (called endergonic), whereas a negative $\Delta G^{0\prime}$ needs no energy input and actually releases excess energy (called exergonic).
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 3.4

11) Explain what an enzyme must accomplish to catalyze a specific reaction.
Answer: Answers will vary, but the focus of the answer should be on overcoming the required activation energy.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.5

12) A beer-making microbiologist noticed that no matter how long the brewing process went, 3% alcohol was the maximum produced. Hypothesize what is causing this low level of alcohol in reference to the brewer's recipe and recommend how a higher alcohol yield could be achieved. Ethanol is toxic at high concentrations, but ignore this factor to focus on microbial metabolism.
Answer: Answers will vary but one explanation is a low substrate concentration resulted in low fermentation to produce ethanol. Providing more carbohydrates such as glucose to the yeast in the recipe for the same growth period would increase fermentation activity and ethanol production. Another explanation is that there may be too much oxygen introduced during the brewing process, which would result in the complete oxidation of glucose instead of fermentation to ethanol. The brewer would need to take more precautions to exclude oxygen during brewing.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.8

13) Explain why the amount of energy released in a redox reaction depends on the nature of both the electron donor and the electron acceptor.
Answer: Answers should emphasize that energy does not come from specific molecules but rather from the difference in reduction potential between two molecules. For example, assigning arbitrary values and subtracting them from one another by comparing two different electron acceptors to one donor would indicate differences in energy for an electron acceptor. In a similar way, this could also be shown to mathematically explain electron donors having an equal role in determining $\Delta E0'$.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 3.6

14) Consider a pizza dough made by vigorously mixing to form gluten and evenly disperse the ingredients such as baker′s yeast (*Saccharomyces cerevisiae*). Predict the metabolic differences yeast would have in a thinly flattened dough and in a spherical dough ball.
Answer: A flattened dough would have higher surface area and more oxygen exposure to support aerobic respiration of *S. cerevisiae*. The dough ball on the other hand would initially have aerobic metabolism of *S. cerevisiae* due to the mixing. Once oxygen is depleted from respiration the yeast would begin anaerobic fermentation, especially in the center of the dough ball while the surface of the dough ball could still support aerobic growth if not enclosed in a container.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 3.8